CLOUD-99: Best Practice Summary

> **Series:** CLOUD — Cloud Provider Integrations | **Notebook:** 99 | **Created:** March 2026 | **Last Updated:** 10/25/2026

Overview

This notebook distills every actionable best practice from the CLOUD series (CLOUD-01 through CLOUD-08) into definitive, priority-ranked tables. Each entry specifies the exact setting, value, or action to take. No hedging — these are the recommended configurations.

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1. Connection & Authentication

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Use Clouds app direct connections for SaaS	**Clouds app > Add connection** — no ActiveGate required	Critical	CLOUD-01
2	Reserve classic ActiveGate polling for Managed only	ActiveGate-based polling for Dynatrace Managed or strict network isolation only	Critical	CLOUD-01

#	Best Practice	Recommended Setting/Value	Priority	Source
3	AWS: Use IAM role-based auth (STS AssumeRole)	IAM Role with trust policy to Dynatrace AWS account; never use long-lived access keys in production	Critical	CLOUD-02
4	AWS: Use the Dynatrace-generated scoped read-only policy	Use the Dynatrace-generated scoped read-only policy from the CloudFormation template (or a least-privilege policy derived from it); never `AdministratorAccess`	Critical	CLOUD-02
5	Azure: Use Azure Native Dynatrace Service	Deploy from Azure Marketplace for zero-infrastructure setup with unified billing	Recommended	CLOUD-05
6	Azure: Use managed identity only for a classic ActiveGate on an Azure VM	For Clouds-app connections the federated identity credential (row 8) is the recommended method; managed identity applies only to the classic ActiveGate path	Recommended	CLOUD-05
7	Azure: Assign `Monitoring Reader` role at subscription scope	Scope: subscription or management group; role: `Monitoring Reader` (the role the Azure connection docs assign); `Reader` only for the classic integration	Critical	CLOUD-05
8	Azure: Use a federated identity credential (non-native)	Use a federated identity credential for Clouds-app Azure connections; client secret only where federation is unavailable (keep expiry < 12 months)	Recommended	CLOUD-05
9	GCP: No service account keys	Clouds-app connection: impersonation — grant the Dynatrace principal `roles/iam.serviceAccountTokenCreator` on a read-only service account; `dynatrace-gcp-monitor`: Workload Identity Federation for GKE with its generated custom roles	Critical	CLOUD-06
10	GCP: Use dedicated project for monitoring resources	Isolate Dynatrace service accounts, Pub/Sub topics, and billing in a separate GCP project	Recommended	CLOUD-06
11	GCP: Choose the connection model deliberately	Clouds-app GCP connection is Preview; `dynatrace-gcp-monitor` on GKE is in maintenance mode but is the working production path. It polls metrics and pulls logs from Pub/Sub (Pub/Sub carries logs only)	Recommended	CLOUD-06

2. AWS Integration

#	Best Practice	Recommended Setting/Value	Priority	Source
12	Enable services in tiers, not all at once	**Tier 1 (always):** EC2, ELB, RDS, Lambda. **Tier 2:** S3, SQS, SNS, DynamoDB, ECS/EKS. **Tier 3:** CloudFront, Route 53, Kinesis, Step Functions. **Tier 4:** SageMaker, Bedrock, IoT	Critical	CLOUD-02
13	Use tag-based resource filtering	Tag key: `dynatrace-monitored`, value: `true`; only monitored resources are discovered	Critical	CLOUD-02
14	Scope monitoring to active regions only	Disable regions with no workloads in the Clouds app connection settings	Recommended	CLOUD-02
15	Account for CloudWatch metric delays in alert windows	Standard metrics: 5–10 min delay. Set alert evaluation window ≥ 15 min for cloud metrics	Critical	CLOUD-02
16	Enable CloudWatch detailed monitoring for critical EC2	Detailed monitoring: 1-min granularity (additional cost); standard: 5-min	Recommended	CLOUD-02
17	Combine cloud integration with OneAgent on compute	Cloud integration = cloud context (tags, AZ, instance type); OneAgent = deep observability (traces, code-level). Use both	Critical	CLOUD-01
18	Review CloudWatch billing dashboard monthly	Correlate API cost spikes with Dynatrace service additions	Recommended	CLOUD-02
18a	Promote AWS tags to primary tags in two steps, plus OneAgent	AWS tag rule (target Cloud tag) at account scope → OpenPipeline primary Grail tag rule with sources `aws.tags.<key>` and `` `tags:aws`[<key>] ``, condition `isNull(primary_tags.<key>)`; Host/Process property rule at host-group scope for OneAgent on EC2 (SaaS 1.348+, new cloud connections)	Critical	CLOUD-02

> **Sources:** AWS integration (DT docs). **Derived:** the recommended settings in this table are this series' positions, distilled from CLOUD-01 through CLOUD-08 — each notebook carries the per-claim sources.

3. Azure Integration

#	Best Practice	Recommended Setting/Value	Priority	Source
19	Map Azure resource groups to Segments (views) and IAM policies (access)	Segments filter by `azure.resource.group` / `azure.subscription` but grant no access — restrict access with IAM policies or bucket permissions	Critical	CLOUD-05
20	Propagate Azure tags to Dynatrace	Tags must flow through automatically; verify `environment`, `team`, `cost-center` appear in Dynatrace	Recommended	CLOUD-05
21	Use naming convention for resource groups	Pattern: `rg-<application>-<environment>-<region>` (e.g., `rg-myapp-prod-eastus`)	Recommended	CLOUD-05
22	Forward Azure logs through Diagnostic Settings → Event Hubs	Diagnostic settings stream activity, Entra ID and resource logs to regional Event Hubs that Dynatrace pulls from (no function code to host); the Azure Native service can create the settings for you	Recommended	CLOUD-05
22a	Promote Azure tags the same way	Azure tag rule → primary Grail tag rule with `azure.tags.<key>` and `` `tags:azure`[<key>] ``; confirm the Azure tag rule type exists in your tenant first	Recommended	CLOUD-05

> **Sources:** Azure Native Dynatrace Service (DT docs). **Derived:** the recommended settings in this table are this series' positions, distilled from CLOUD-01 through CLOUD-08 — each notebook carries the per-claim sources.

4. GCP Integration

#	Best Practice	Recommended Setting/Value	Priority	Source
23	Use Workload Identity for GKE pods	Replace service account JSON keys with Workload Identity for DynaKube and integration pods	Critical	CLOUD-06
24	Tag all GCP resources with standard labels	Required labels: `env`, `team`, `service`, `cost-center`; propagate as Dynatrace tags	Recommended	CLOUD-06
25	Scope by the `gcp.project.id` field	The project arrives as a field on nodes, logs and metrics — build segments and filters on it instead of a hand-propagated tag	Recommended	CLOUD-06
26	Create Dynatrace Segments per GCP project	One Segment per project for cost attribution and scoped dashboards (segments filter; access is an IAM-policy job)	Recommended	CLOUD-06

#	Best Practice	Recommended Setting/Value	Priority	Source
27	Monitor Pub/Sub message volume for cost control	Filter logs at Cloud Logging export to reduce Pub/Sub throughput and costs	Recommended	CLOUD-06
28	Use Cloud Monitoring metrics scope for multi-project	Aggregate metrics from multiple projects into a single metrics scope rather than separate integrations	Optional	CLOUD-06
28a	Promote GCP labels and tags the same way	GCP label / GCP tag rule → primary Grail tag rule with `gcp.labels.<key>` and `` `tags:gcp_labels`[<key>] ``	Recommended	CLOUD-06

> **Sources:** Google Cloud integration (DT docs). **Derived:** the recommended settings in this table are this series' positions, distilled from CLOUD-01 through CLOUD-08 — each notebook carries the per-claim sources.

5. Kubernetes (EKS / AKS / GKE)

#	Best Practice	Recommended Setting/Value	Priority	Source
29	Deploy DynaKube in CloudNativeFullStack mode for standard node pools	`spec.oneAgent.cloudNativeFullStack` — full agent injection + infrastructure monitoring	Critical	CLOUD-03
30	Use ApplicationMonitoring mode for Fargate; check mode support on GKE Autopilot	`spec.oneAgent.applicationMonitoring`; on GKE Autopilot the Operator's chart creates the allowlist for its CSI driver — confirm per-mode support for your Autopilot version; for Fargate, install the Operator with the **Without CSI driver** variant (`useCSIDriver` is not a DynaKube `v1beta5` / `v1beta6` field)	Critical	CLOUD-03, CLOUD-06
31	Add tolerations for tainted node groups	`tolerations: [{effect: NoSchedule, key: node-role, operator: Exists}]` to ensure OneAgent runs on all nodes	Critical	CLOUD-03
32	Enable kubernetes-monitoring capability on ActiveGate	`spec.activeGate.capabilities: [kubernetes-monitoring, routing]`	Critical	CLOUD-03
33	Use nodeSelector to exclude Windows nodes	`nodeSelector: {kubernetes.io/os: linux}` in DynaKube spec	Recommended	CLOUD-03
34	Enforce namespace labels for cost allocation	Require `team`, `cost-center`, `environment` labels on all namespaces	Recommended	CLOUD-03
35	Set resource requests and limits on all pods	Accurate requests enable fair cost attribution and prevent resource contention	Recommended	CLOUD-03

#	Best Practice	Recommended Setting/Value	Priority	Source
36	Use Dynatrace for workload monitoring, Container Insights for control plane	Keep CloudWatch Container Insights / Azure Monitor for control plane logs; Dynatrace for distributed tracing and AI alerting	Recommended	CLOUD-03
37	Use IRSA (AWS) or Workload Identity (GCP) for pod IAM	Never mount service account JSON keys into pods; use cloud-native identity federation	Critical	CLOUD-03, CLOUD-06

6. Serverless Monitoring

#	Best Practice	Recommended Setting/Value	Priority	Source
38	Install Dynatrace Lambda Layer on all Lambda functions	Enables distributed tracing, code-level visibility, and direct log collection	Critical	CLOUD-04
39	Use meaningful Lambda function names	Avoid auto-generated names; use `<service>-<function>-<env>` pattern for identification	Recommended	CLOUD-04
40	Monitor concurrency headroom	Alert when `cloud.aws.lambda.ConcurrentExecutions.By.FunctionName` (Metric Streams) / `dt.cloud.aws.lambda.conc_executions` (polling) exceeds 80% of account limit (default: 1,000/region)	Critical	CLOUD-04
41	Track cold start ratio via spans	Query `faas.coldstart == true` from spans; high ratio indicates need for provisioned concurrency or SnapStart	Recommended	CLOUD-04
42	Use error-to-invocation ratio, not raw error count	`error_pct = errors / invocations * 100`; alert threshold: > 1% for critical functions	Critical	CLOUD-04
43	Correlate API Gateway latency with Lambda duration	End-to-end latency = API GW overhead + Lambda duration; monitor both	Recommended	CLOUD-04

#	Best Practice	Recommended Setting/Value	Priority	Source
44	Monitor DynamoDB call duration from Lambda spans	Filter: `span.kind == "client"`, `db.system == "dynamodb"`; alert on p95 > baseline	Recommended	CLOUD-04
45	Set alert evaluation window longer than metric delay	Cloud metric delay: 5–10 min. Minimum evaluation window for Lambda alerts: 15 min	Critical	CLOUD-02
45a	Tag Lambda functions with DT_TAGS	`DT_TAGS="primary_tags.team=… aws.account.id=… aws.region=…"` — the function does not load ingest enrichment rules	Critical	CLOUD-04

7. Log Ingestion

#	Best Practice	Recommended Setting/Value	Priority	Source
46	Use Amazon Data Firehose for CloudWatch log forwarding	Fully managed, auto-scaling, no custom code. Buffer: 1 MB or 60 seconds. Enable GZIP compression	Critical	CLOUD-07
47	Migrate the legacy **CloudWatch-subscription** Lambda forwarder to Firehose	`dynatrace-aws-log-forwarder` — the CloudWatch Subscription Filter → Lambda → Dynatrace API path — is **deprecated**; do not use it for new *CloudWatch* log forwarding. **Scope of this row:** it retires one named component for one source. It is **not** a prohibition on Lambda-based log forwarding in general — see 47b	Critical	CLOUD-07
47b	For logs already in **S3**, use the S3 direct-ingestion forwarder — a *different* component for a *different* source, and supported	Available from **SaaS 1.344** (**staged tenant rollout** from 07/29/2026 — verify it has reached your tenant): a **Dynatrace-maintained** Lambda function in your centralized logging account, triggered by S3 event notifications, deployed by a **single CloudFormation stack**, with records **linked to their Smartscape entity** when the producing account has an active AWS connection. Until it arrives, the Dynatrace-published `dynatrace-aws-s3-log-forwarder` (not the archived `dynatrace-aws-log-forwarder`) remains the working path — CLOUD-07 §3	Recommended	CLOUD-07
48	Use Lambda Layer log collection for Lambda functions	When Lambda Layer is deployed for tracing, use its built-in log collection instead of Firehose for Lambda logs	Recommended	CLOUD-07

#	Best Practice	Recommended Setting/Value	Priority	Source
49	Apply subscription filter patterns at CloudWatch level	Filter pattern examples: `?"ERROR" ?"WARN" ?"CRITICAL"` for Lambda; `""` (all) for application services	Critical	CLOUD-07
50	Drop debug/trace logs at source	Pre-filter: exclude DEBUG and TRACE in CloudWatch subscription filters. Estimated savings: 40–60%	Critical	CLOUD-07
51	Filter health check log lines at source	Remove `/health`, `/ready`, `/live` endpoint noise before forwarding. Savings: 10–30%	Recommended	CLOUD-07
52	Use OpenPipeline for enrichment, routing, and fine-grained filtering	Add `cloud.provider`, route by log group to specific Grail buckets, drop remaining noise	Recommended	CLOUD-07
53	Route logs to purpose-specific Grail buckets	Route `/aws/lambda/*` → `lambda_logs`, `/ecs/*` → `application_logs`, `/aws/rds/*` → `database_logs`	Recommended	CLOUD-07
54	Enable S3 backup on Firehose delivery stream	Backup failed or all records to S3 for disaster recovery and compliance	Recommended	CLOUD-07
55	Azure: One diagnostic setting per resource, regional Event Hubs	Event Hubs must be in the resource's region; each resource allows at most five diagnostic settings	Recommended	CLOUD-05
56	GCP: Aggregated Log Router sink → Pub/Sub	One folder/organization sink with `includeChildren`; exclusion filters before Pub/Sub; grant the writer identity `roles/pubsub.publisher`	Recommended	CLOUD-06

8. Cost Optimization

#	Best Practice	Recommended Setting/Value	Priority	Source
57	Disable unused cloud service monitoring	Only enable AWS/Azure/GCP services your applications depend on; disable the rest	Critical	CLOUD-02
58	Tag-based filtering is the #1 cost lever	Monitor only resources tagged `dynatrace-monitored: true`; reduces both DPS and cloud API costs	Critical	CLOUD-02
59	Disable individual metrics you do not need	Per-service metric selection in Clouds app or Settings UI	Recommended	CLOUD-02
60	Consider CloudWatch Metric Streams at scale	Streaming can be cheaper than polling for >100 resources with frequent metric retrieval	Optional	CLOUD-02
61	Set bucket-level retention policies in Grail	Low-value logs: 7–14 days. Standard logs: 35 days. Compliance logs: 365+ days	Critical	CLOUD-07

#	Best Practice	Recommended Setting/Value	Priority	Source
62	Monitor log volume weekly	Query: `fetch logs, from:-7d | makeTimeseries log_count = count(), interval:1h` — investigate spikes immediately	Recommended	CLOUD-07
63	Rightsize overprovisioned instances	Alert threshold: CPU < 10% sustained over 24h. Action: downsize or terminate	Recommended	CLOUD-08
64	Identify zero-traffic services for decommission	Query services with no spans/requests over 7 days; flag as decommission candidates	Optional	CLOUD-08
65	Enforce K8s resource requests/limits	Pods without requests/limits cause uncontrolled resource usage and inaccurate cost allocation	Recommended	CLOUD-03, CLOUD-08

9. Multi-Cloud Governance

#	Best Practice	Recommended Setting/Value	Priority	Source
66	Apply consistent tag schema across all providers	Required tags: `cloud-provider`, `environment`, `team`, `application`, `cost-center`, `region`	Critical	CLOUD-08
67	Normalize region names across providers	Use normalized values: `us-east`, `eu-west`, `ap-south` — not provider-specific names	Recommended	CLOUD-08
68	Create Dynatrace Segments per cloud provider + environment	One Segment per combination (e.g., `AWS-Production`, `Azure-Staging`) for scoped views; pair with IAM policies for access control, which segments do not provide	Critical	CLOUD-08
69	Enforce tagging compliance with automation	Reject or alert on untagged cloud resources; query: `fetch dt.entity.host | filter isNull(tags)`	Critical	CLOUD-08
70	Define SLOs at the service level, not infrastructure level	SLOs are cloud-agnostic: measure user experience (latency, error rate, availability)	Recommended	CLOUD-08
71	Route alerts by team, not by cloud provider	Dynatrace Workflows route to the application owner regardless of which cloud has the issue	Critical	CLOUD-08
72	Use unified runbooks referencing DQL queries	Runbooks should use Dynatrace DQL, not provider-specific CLI tools	Recommended	CLOUD-08
73	Conduct monthly utilization reviews across all providers	Review underutilized hosts (CPU < 10%), untagged resources, and zero-traffic services	Recommended	CLOUD-08

10. Alerting & Observability

#	Best Practice	Recommended Setting/Value	Priority	Source
74	Use Dynatrace Intelligence for automatic cross-cloud anomaly detection	Enable Anomaly Detection on all monitored entities; do not rely solely on static thresholds	Critical	CLOUD-08
75	Set alert evaluation windows ≥ 15 min for cloud metrics	Cloud metric delay (planning estimates, CLOUD-02): CloudWatch standard 5–10 min, ActiveGate polling 5–15 min. A 1-min evaluation window on 5-min metrics produces false negatives	Critical	CLOUD-02
76	Use OneAgent metrics for sub-minute alerting	For alerts requiring < 5 min evaluation, use `dt.host.*` or `dt.process.*` metrics from OneAgent, not cloud metrics	Critical	CLOUD-02
77	Combine cloud integration with OneAgent on all compute	Cloud integration: infrastructure context. OneAgent: processes, traces, code-level diagnostics. Both required for full-stack visibility	Critical	CLOUD-01
78	Build unified health dashboards showing all providers	Components: host CPU (all providers), active detected problems, service error rates, log error trends, K8s container metrics	Recommended	CLOUD-08
79	Deduplicate alerts using Dynatrace Intelligence correlation	Dynatrace Intelligence automatically correlates related issues across providers; do not create redundant static alerts	Recommended	CLOUD-08
80	Forward control plane logs to Dynatrace for unified analysis	EKS: Container Insights logs. AKS: control-plane resource logs via diagnostic settings (prefer `kube-audit-admin` over `kube-audit` for cost). GKE: audit and control-plane logs via a Log Router sink → Pub/Sub. Forward all to Grail for cross-platform analysis	Recommended	CLOUD-03, CLOUD-06

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